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Skill 21.1 Exercise 1

The two code segments below both open, read, and close the file *document.txt*.

Segment 1	**Segment 2**

`my_doc = open('document.txt')`
`text = my_doc.read()`
`my_doc.close()`
`print(text)`
	`with open('document.txt') as my_doc:`
	`    text = my_doc.read()`
	
	`print(text)`

What are the advantages of *Segment 2* over *Segment 1*?

If **any error occurs** between open() and close() , my_doc.close() will **never run**. Just a simple typo for example.

a with block guarantees the file is closed properly, even if something goes wrong.

Skill 21.2 Exercise 1

Refer the text which is stored in *some_lines.txt*

some_lines.txt

1. How many lines do we write on the daily
2. Many money, we write many many many
3. How many lines do you write on the daily
4. Say you say many money, you write many many many

Use a *with* statement to create a file object pointing to the file *some_lines.txt* shown to the right. Store that file object in the variable *lines_doc*.

with open("some_lines.txt") as lines_doc:

Assume what you write above is correct, what is printed for each of the following statements.

```
print(lines_doc)
print(lines_doc.readlines())
```

The file object

A list of the lines

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Iterate through each of the lines in lines_doc.readlines() using a for loop. Inside the for loop print out each line of *some_lines.txt*.
all_lines = lines_doc.readlines() for line in all_lines: print(line)

Skill 21.2 Exercise 2
Refer the text which is stored in *some_lines.txt*
some_lines.txt 1. How many lines do we write on the daily 2. Many money, we write many many many 3. How many lines do you write on the daily 4. Say you say many money, you write many many many
Use a *with* statement to create a file object pointing to the file *some_lines.txt* shown to the right. Store that file object in the variable *lines_doc*.
with open("some_lines.txt") as lines_doc:

Assume what you write above is correct, what is printed for each of the following statements.
 	
`lines_doc.readline()`
`lines_doc.readline()`
`lines_doc.readline()`
`print(lines_doc.readline())`

4. Say you say many money, you write many many many

Skill 21.3 Exercise 1
Create a file object for the file *bad_bands.txt* using the *open()* function with the *w* argument. Assign this object to the temporary variable *bad_bands_doc*. Use the *bad_bands_doc.write()* method to add the name of a musical group you dislike to the document *bad_bands.txt*.
with open('bad_bands.txt', 'w') as bad_bands_doc: bad_bands_doc.write("Twisted Sister")

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Skill 21.4 Exercise 1

In a previous exercise, we created a file called *bad_bands.txt.*

Write two additional bad bands to this file by using the append argument, *'a'*. Each band should be added on a separate line.

```
with open('bad_bands.txt', 'a') as bad_bands_doc:
  bad_bands_doc.write("\nMotley Crew")
  bad_bands_doc.write("\nRATT")
```

Skill 21.5 Exercise 1

Refer to the csv file below,

users.csv

```
Name,Username,Email
Roger Smith,rsmith,wigginsryan@yahoo.com
Michelle Beck,mlbeck,hcosta@hotmail.com
Ashley Barker,a_bark_x,a_bark_x@turner.com
Lynn Gonzales,goodmanjames,lynniegonz@hotmail.com
Jennifer Chase,chasej,jchase@ramirez.com
Charles Hoover,choover,choover89@yahoo.com
Adrian Evans,adevans,adevans98@yahoo.com
Susan Walter,susan82,swilliams@yahoo.com
Stephanie King,stephanieking,sking@morris-tyler.com
Erika Miller,jessica32,ejmiller79@yahoo.com
```

In the space below do the following,

- Import the csv module
- Create an empty list called *user_emails*
- Open up the file *users.csv* in the temporary variable *users_file*.
- Use the *csv.DictReader* method to read the contents of *users_file* into a new variable called *users_dict*.
- Use a loop to iterate over each *users_dict* and store the email for each user in *user_emails*

```
import csv

user_emails = []
with open('users.csv', newline='') as users_file:
  user_dict = csv.DictReader(users_file)  #turns the users_file into rows of dictionaries
  for row in user_dict:
    user_emails.append(row["Email"])
```

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Skill 21.6 Exercise 1

Refer to the csv file below,

books.csv

```
Author@ISBN@Title
Lauren Murray@978-0-12-995015-8@"Enviornment Call, Amount Later Page Country"
Micheal Jones@978-1-78110-100-1@Rate Security Full
Alexander Carr@978-0-315-25137-3@Still Response Size
Michael Williams@978-0-388-70665-7@Position Result Five
Kathleen Ferguson@978-1-75098-721-6@Country Week Receive And Sign
Sarah Dorsey@978-1-06-483628-6@Audience Truth Small
Mary Middleton@978-0-7419-8114-1@Travel: Special Offer Near Allow Goal
William Todd@978-1-4457-0480-7@Money Exactly Drop Teach
Joan Martin@978-0-657-61030-2@Theory Do Half Change
Gary Roman@978-1-5039-7539-2@Bill Serve Pull Industry South Job
```

In the space below do the following,

- Import the csv module
- Open up the file *books.csv* in the temporary variable *books_file*.
- Use the *csv.DictReader* method to read the contents of *books_file* into a new variable called *books_dict* (what delimiter is being used here?).
- Create a list called *isbn_list*
- Use a loop to iterate over each *books_dict* and store the ISBN for each book in *isbn_list*

```
import csv
with open("books.csv") as books_file:
 books_dict = csv.DictReader(books_file, delimiter = "@")
 isbn_list = []
 for book in books_dict:
  isbn_list.append(book["ISBN"])
 print(isbn_list)
```

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Skill 21.7 Exercise 1

Refer to the list of access log dictionaries below,

```
access_log = [{'time': '08:39:37', 'limit': 844404, 'address': '1.227.124.181'},
{'time': '13:13:35', 'limit': 543871, 'address': '198.51.139.193'}, {'time':
'19:40:45', 'limit': 3021, 'address': '172.1.254.208'}, {'time': '18:57:16',
'limit': 67031769, 'address': '172.58.247.219'}, {'time': '21:17:13', 'limit':
9083, 'address': '124.144.20.113'}, {'time': '23:34:17', 'limit': 65913, 'address':
'203.236.149.220'}, {'time': '13:58:05', 'limit': 1541474, 'address':
'192.52.206.76'}, {'time': '10:52:00', 'limit': 11465607, 'address':
'104.47.149.93'}, {'time': '14:56:12', 'limit': 109, 'address': '192.31.185.7'},
{'time': '18:56:35', 'limit': 6207, 'address': '2.228.164.197'}]

fields = ['time', 'address', 'limit']
```

In the space below, do the following

- Import the csv module
- Open up the file *logger.csv* in the temporary variable *logger_csv*. Don't forget to open the file in write-mode.
- Create a *csv.DictWriter* instance called *log_writer*. Pass *logger_csv* as the first argument and then *fields* as a keyword argument to the keyword *fieldnames*.
- Write the header to *log_writer* using the *.writeheader()* method.
- Iterate through the *access_log* list and add each element to the CSV using *log_writer.writerow()*.

```
with open("logger.csv", "w") as logger_csv:
 fields = ['time', 'limit', 'address']
 log_writer = csv.DictWriter(logger_csv, fieldnames = fields)
 log_writer.writeheader()
 for log in access_log:
  log_writer.writerow()
```

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Skill 21.8 Exercise 1

Refer to the json file below,

message.json
{'text': "Now that's JSON!", 'secret text': "Now that's some _serious_ JSON!"}

In the space below, do the following,

- Import the json module
- Open up the file *message.json* and save the file object to the variable *message_json*.
- Pass the JSON file object as an argument to *json.load()* and save the resulting Python dictionary as *message*.
- Print out *message['text']*.

```
import json
with open('message.json') as message_json:
  message = json.load(message_json)
  print(message[0]["text"])
```

What is the output from the print statement above?

Now that's JSON!

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Skill 21.9 Exercise 1

Refer to the dictionary below,

```
data_payload = {'interesting message': 'What is JSON? A web application\'s little
pile of secrets.',
   'follow up': 'But enough talk!'}
```

In the space below do the following,

- Import the json module
- Open a new file object in the variable *data_json*. The filename should be *data.json* and the file should be opened in write-mode.
- Call *json.dump()* with *data_payload* and *data_json* to convert our data to JSON and then save it to the file *data.json*.

```
import json

with open("data.json", "w") as data_json:
   json.dump(data_payload, data_json)
```